

# Lecture 2 - Internet Design Principles Quiz

## ANS

**1. Which of the following best describes "Decentralized control" as an Internet design principle?**

- A) A single central server controls all network traffic
- B) Each network device (e.g., router) runs independently without a central mastermind
- C) The government manages all network operations
- D) All routers must communicate with a master node before forwarding packets

**ANS: B**

Decentralized control means each network device operates autonomously without requiring approval from a central authority. This allows the Internet to function reliably even if individual components fail.

**2. What is the "narrow waist" of the Internet?**

- A) The slowest part of the network connection
- B) A physical constraint on cable diameter
- C) IP (Internet Protocol) being the only protocol at Layer 3
- D) The maximum bandwidth available on a single link

**ANS: C**

The narrow waist refers to IP being the universal protocol at Layer 3. All hosts and routers understand IP, which unifies the Internet and enables federation among different network operators.

**3. How does demultiplexing at Layer 3 work?**

- A) By encrypting packets so only the intended recipient can read them
- B) By adding a header field that tells the receiving end what the next higher-layer protocol is
- C) By splitting large packets into smaller pieces
- D) By routing packets along multiple paths simultaneously

**ANS: B**

Demultiplexing uses header fields to indicate which Layer 4 protocol (e.g., TCP or UDP) should handle the packet. This allows the IP code to forward the packet to the correct higher-layer protocol handler.

**4. What is a "logical port" in networking?**

- A) The physical cable connector on a network interface card
- B) A number identifying an application, existing in software
- C) A dedicated circuit established between two hosts
- D) The maximum speed a link can transmit data

**ANS: B**

A logical port is a software-based number (0-65535) that uniquely identifies an application or connection on a computer. This is distinct from a physical port, which is the hardware connector where cables are plugged in.

**5. Why do well-known ports (like port 80 for HTTP) have fixed numbers?**

- A) To reduce the size of packets
- B) To increase network security
- C) So that clients can find the server without pre-arrangement
- D) To limit the number of simultaneous connections

**ANS: C**

Public servers like web servers must use fixed, well-known port numbers so that clients can connect to them without prior negotiation. Private clients can use randomly-generated port numbers.

**6. What does the End-to-End Principle state?**

- A) Reliability must be implemented at the end hosts, not solely in the network
- B) All packets must travel the same path from source to destination
- C) The network should guarantee delivery of every packet
- D) End hosts should trust routers to handle all error detection

**ANS: A**

The End-to-End Principle argues that certain application features like reliability must be implemented at end hosts for correctness. Relying only on network routers for reliability is insufficient if the router implementation is buggy.

**7. In the simple reliability protocol example, why is Solution 1 (reliability in the network) insufficient?**

- A) It requires too much bandwidth
- B) Routers might be buggy, and the end host has no way to verify correctness
- C) It's incompatible with IP
- D) It increases packet size

**ANS: B**

Solution 1 fails because if a router reports success when it's actually buggy, the end host has no independent way to verify the claim. The end host must perform its own reliability checks for correctness.

**8. The End-to-End Principle can be relaxed for what reason?**

- A) To reduce the complexity of end host implementations
- B) As a performance optimization, provided end-to-end checks still occur
- C) To simplify router implementations
- D) To reduce network congestion

**ANS: B**

While the End-to-End Principle is a design guideline, it can be relaxed for performance optimization (e.g., routers sending duplicate packets on lossy links). However, this must supplement, not replace, end-to-end reliability checks.

**9. What is Statistical Multiplexing?**

- A) Sending the same packet multiple times to ensure delivery
- B) Dynamically allocating resources to users based on their current demand
- C) Dividing network resources equally among all users at all times
- D) Multiplying the number of routers to handle more traffic

**ANS: B**

Statistical multiplexing allocates network resources dynamically based on current usage patterns. This is more efficient than static allocation because peak demand doesn't occur simultaneously for all users.

**10. With statistical multiplexing, why is dynamic allocation more efficient than fixed allocation?**

- A) It reduces the number of packets
- B) It uses less powerful hardware
- C) Peak of aggregate demand is less than the aggregate of peak demands
- D) It eliminates the need for routers

**ANS: C**

When Alice peaks in the morning and Bob peaks at night, dynamic allocation needs only 12 units total capacity (10+2 in morning, 2+10 at night). Fixed allocation would require 20 units to satisfy both at their peak times.

**11. What is Circuit Switching?**

- A) Randomly selecting a path for each packet
- B) Explicitly reserving resources at connection start and releasing them at connection end
- C) Sharing network bandwidth equally among all active flows
- D) Switching between multiple network providers

**ANS: B**

Circuit switching requires end hosts to explicitly request and reserve network resources (e.g., 5 Mbps) before sending data. Routers establish a dedicated circuit for the connection, and resources are freed when the connection ends.

**12. Which switching approach is more efficient for bursty traffic (e.g., web browsing)?**

- A) Circuit switching, because it reserves capacity
- B) Packet switching, because it doesn't waste reserved capacity on idle periods
- C) Both are equally efficient
- D) Neither is efficient for bursty traffic

**ANS: B**

Packet switching is more efficient for bursty traffic because resources are allocated per-packet, not reserved for the entire connection. With circuit switching, reserved capacity sits idle when demand drops, wasting resources.

**13. Why is Packet Switching better than Circuit Switching at handling network failures?**

- A) Packet switching uses encryption to protect against failures
- B) Packet switching automatically reroutes packets; end hosts need not do anything extra
- C) Packet switching prevents routers from failing
- D) Circuit switching requires immediate manual intervention

**ANS: B**

With packet switching, routers can automatically reroute packets along alternate paths without end-host involvement. Circuit switching requires end hosts to detect failure, tear down reservations, and request new reservations—potentially for millions of flows simultaneously.

**14. Which of the following is NOT an advantage of Packet Switching over Circuit Switching?**

- A) More efficient for bursty traffic
- B) Better performance predictability for applications
- C) Simpler implementation
- D) Better handling of network failures

**ANS: B**

Packet switching's main disadvantage is that it offers less predictability. Applications don't get resource guarantees like they do with circuit switching, making performance less predictable. All other options (A, C, D) are advantages of packet switching.